Predictive Stochastic Modeling for Financial Markets

Revised edition

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This revised edition of the undergraduate thesis that received the Adam Smith Award evaluates bear-market forecasts in real time and adds a broad set of robustness experiments. All estimates, tables and figures are produced by a single reproducible pipeline. The award version (February 2025), its R code and the code for this edition are available at github.com/jadenfix/Predictive-Stochastic-Modeling-for-Financial-Markets-.

Abstract

Can market and macroeconomic indicators forecast bear markets in the S&P 500? I date 21 bear markets in the real S&P 500 since 1900 with a threshold rule, describe monthly returns with a two-regime Markov-switching model, and evaluate probability forecasts made every month from January 1960 to December 2023, with every model re-estimated on data that were available at the forecast origin. Returns alternate between a calm bull regime (mean 0.84% a month, expected duration 17.9 months) and a volatile bear regime (mean −1.81%, expected duration 5.2 months). Out of sample, a logit on 6 state variables reduces the one-month Brier score by 20.8% relative to the unconditional bear frequency. Knowing only whether a confirmed bear market is under way achieves 18.9%, and none of the 20 model-horizon pairs beats this persistence benchmark at the 5% level. Skill declines with the horizon; at twelve months no model is reliably better than the base rate. The conclusions hold across alternative dating thresholds and nominal prices, five regime specifications, rolling estimation windows, subperiods, 5 additional predictors, forecast combination and block-bootstrap inference. The one predictor that beats persistence is the drawdown from the twelve-month high at one month, which recognizes a decline already in progress slightly earlier than the dating rule. Bear markets are partly identifiable in real time; they are not meaningfully predictable months in advance with these data.

1 Introduction

Bear markets destroy a large share of household and institutional wealth in a short time, and whether they can be anticipated is a direct test of how much public information stock prices already contain. Two literatures bear on the question. One finds that equity returns alternate between persistent bull and bear regimes (Hamilton, 1989; Maheu and McCurdy, 2000; Ang and Timmermann, 2012). The other finds that valuation ratios and macroeconomic conditions carry information about future returns and downturns (Campbell and Shiller, 1988; Fama and French, 1988; Chen, 2009). If regimes are persistent and their onset is related to observable conditions, bear markets should be at least partly forecastable.

This paper asks how much of that forecastability survives a strict real-time test. A forecast may use only information that was published when it was made, every model is re-estimated at every forecast origin, and accuracy is measured out of sample against two baselines: the unconditional frequency of bear markets and the persistence of the current, observable market state. The second baseline is the demanding one. A bear market dated from its peak is only recognized once prices have fallen far enough, so a model that merely notices an ongoing decline will look skillful against the base rate without telling an investor anything new.

The analysis has four parts. First, I date bear markets with an explicit threshold rule in the spirit of Lunde and Timmermann (2004), so that the forecast target is an observable event rather than a model output. Second, I estimate two-regime Markov-switching models of real returns and use their filtered probabilities as one forecast. Third, I estimate logit and probit models, the natural tools for a binary outcome (Estrella and Mishkin, 1998; Chen, 2009), on valuation, inflation, interest-rate and market-state predictors, and evaluate their real-time forecasts over 1960:02-2023:12 at horizons of 1, 3, 6, 12 months. Fourth, I subject the results to a robustness program that varies the dating rule, the regime specification, the estimation window, the evaluation period and the predictor set, and that quantifies sampling uncertainty with a block bootstrap.

The evidence supports three findings. Regime structure in returns is strong and stable: a calm bull regime and a short, volatile bear regime appear in every specification and subsample. Indicators

such as inflation, realized volatility and the dividend yield are associated with bear markets in sample, and real-time models beat the unconditional frequency at horizons of one to three months. Most of that short-horizon skill, however, comes from recognizing a bear market that has already begun. No model significantly improves on persistence, and beyond six months nothing reliably improves on the base rate.

The paper proceeds as follows. Section 2 reviews related work, Section 3 describes the data and the dating rule, Section 4 the models and the real-time design, Section 5 the main results and Section 6 the robustness experiments. Section 7 discusses limitations and Section 8 concludes.

2 Related literature

Regime-switching models in the tradition of Hamilton (1989) describe the stock market as moving between a high-return, low-volatility state and a low-return, high-volatility state. Maheu and McCurdy (2000) identify bull and bear markets this way, and Ang and Timmermann (2012) survey the evidence that regime changes are a pervasive feature of financial markets. Rule-based dating is the main alternative: Pagan and Sossounov (2003) adapt business-cycle turning-point algorithms to equity prices, and Lunde and Timmermann (2004) define bull and bear markets by threshold movements from the latest peak or trough. Chen (2009) and Chen, Chen and Chou (2017) use such dates as the dependent variable in probit models and find that macroeconomic and credit variables help predict bear markets.

On the predictor side, Campbell and Shiller (1988) and Fama and French (1988) document that dividend yields forecast long-horizon returns, while Goetzmann and Jorion (1993), Stambaugh (1999) and Lewellen (1999) show that inference with persistent, price-scaled predictors is delicate. Pontiff and Schall (1998) report related evidence for book-to-market ratios, and Hammerschmid and Lohre (2018) find that return predictability itself varies across regimes. Welch and Goyal (2008) show that many predictors that look strong in sample fail out of sample, and Rapach, Strauss and Zhou (2010) find that combining individual forecasts delivers more reliable out-of-sample gains than any single predictor. This paper applies the out-of-sample standard of these studies to the prediction of bear markets rather than of average returns.

3 Data

3.1 Source and timing

The data are the monthly U.S. stock market series compiled by Robert Shiller (Shiller, 2015), archived in the repository as `one45.csv`. The file covers January 1900 to March 2024; I use observations through December 2023, the last month with complete dividend and earnings data. The S&P Composite price is a monthly average of daily closes, so it is known at the end of each month. I treat the end of month t as the forecast origin and impose publication lags on everything else: the consumer price index enters with a lag of 1 month, and dividends and earnings, which Shiller interpolates from quarterly reports, enter with a lag of 6 months. Real values are obtained by deflating with the CPI available at t. Table 1 lists the predictors, their construction and their publication lags.

The construction differs from the award version in one respect that matters for timing. The archived file contains columns labeled `realpricelag` to `realpricelag4` that hold future rather than past prices (the first matches the following month's real price on 1,490 of 1,490 comparable rows), and the award version's regime model and forecast targets inherited this alignment. This edition

Table 1: Predictors, transformations and publication lags

Predictor	Construction	Units	Lag (months)	Mean	SD
Monthly-change predictors (archived thesis variables)					
CPI inflation, monthly	100 dlog CPI	% per month	1	0.31	0.36
Change in 10-year yield	Monthly change	pp	0	0.00	0.27
Real bond return	100 dlog real bond index	% per month	0 (CPI 1)	0.18	1.99
Real dividend growth	100 dlog real dividends	% per month	6	0.17	0.63
Real total return	Price return plus D/12P, less inflation	% per month	0 (D 6, CPI 1)	0.49	3.65
TR-scaled earnings growth	100 dlog real TR-scaled earnings	% per month	6	0.45	4.84
Change in log TR-CAPE	100 dlog TR price / 10-yr mean earnings	% per month	0 (E 6)	0.05	3.66
State predictors (bear-market forecasting models)					
Log dividend yield	log(D / P)	log points	0 (D 6)	−3.67	0.40
Log CAPE	log(P / 10-yr mean real earnings)	log points	0 (E 6)	2.98	0.43
CPI inflation, 12-month	100 x 12-month dlog CPI	%	1	3.66	2.68
12-month change in 10-year yield	12-month change	pp	0	−0.01	1.13
12-month real price change	100 x 12-month dlog real price	%	0 (CPI 1)	3.07	15.94
12-month realized volatility	sd of monthly real returns x sqrt(12)	% annualized	0 (CPI 1)	11.25	4.93
Additional state predictors (robustness)					
Excess CAPE yield	100 E10/P less (yield less 10-yr inflation)	%	0 (E 6, CPI 1)	3.32	2.37
Log payout ratio	log(D / E)	log points	6	−0.76	0.30
12-month real earnings growth	100 x 12-month dlog real earnings	%	6	2.73	35.31
10-year Treasury yield	Level	%	0	5.81	2.95
Drawdown from 12-month high	100 x log(P / 12-month high)	%	0 (CPI 1)	−6.69	9.86

Notes: Statistics over 1960:01-2023:12. A forecast is made at the end of month t. Lag gives the publication lag applied to the inputs: CPI enters with one month, dividends (D) and earnings (E) with six. Real values are deflated by the CPI available at t. The archived 'realpricelag' columns (future prices) are never used.

never reads those columns, builds every predictor from data dated at or before the forecast origin, and verifies with an automated test that no predictor changes when data after its origin are altered. Appendix A summarizes how the award version's estimates relate to the ones reported here.

3.2 Dating bear markets

I date bear markets on the real price index with a symmetric threshold rule in the spirit of Lunde and Timmermann (2004). Starting from a bull state, the running maximum is a candidate peak; when the index falls 20% below it, the peak is confirmed and a bear market begins in the following month. In the bear state the running minimum is a candidate trough; when the index rises 20% above it, the trough is confirmed and the bear market ends at the trough month. The rule yields 21 bear markets since 1900, 9 of them beginning after 1960 (Table 2, Figure 1). The median bear market lasts 20 months with a median peak-to-trough decline of 34.8%. Bear months account for 32.4% of the full sample and 22.7% of January 1960 to December 2023.

The dated labels are ex post: a peak is only known to have been a peak once the 20% decline has occurred, which took a median of 9 months. The same rule run on data through month t gives the real-time state, which is observable at t, and "vintage" labels for all earlier months. The out-of-sample exercise trains every model on vintage labels only.



Figure 1: Real S&P 500 price index (log scale) with dated bear markets shaded.

Table 2: Dated bear markets in the real S&P 500, 1900-2023

Peak	Trough	Months	Decline (%)	Confirmed	Months to confirm
1901-06	1903-10	28	−33.2	1903-06	24
1906-09	1907-11	14	−43.4	1907-03	6
1911-06	1914-12	42	−34.8	1913-05	23
1915-11	1920-12	61	−62.9	1917-05	18
1929-09	1929-11	2	−34.2	1929-11	2
1930-04	1932-06	26	−76.9	1930-10	6
1932-09	1933-02	5	−20.8	1933-02	5
1933-07	1935-03	20	−30.6	1934-07	12
1937-02	1938-04	14	−45.4	1937-09	7
1938-11	1942-04	41	−47.5	1940-06	19
1946-04	1949-06	38	−42.4	1946-09	5
1956-07	1957-12	17	−20.8	1957-11	16
1961-12	1962-06	6	−23.0	1962-06	6
1968-12	1970-07	19	−35.1	1970-01	13
1973-01	1974-12	23	−53.3	1973-12	11
1976-09	1980-04	43	−30.0	1978-01	16
1980-11	1982-08	21	−29.7	1982-01	14
1987-08	1987-12	4	−27.9	1987-11	3
2000-08	2003-02	30	−46.4	2001-03	7
2007-10	2009-03	17	−51.7	2008-07	9
2021-11	2022-10	11	−25.6	2022-06	7

Notes: A bear market runs from a peak to the following trough when the real index falls at least 20% below the peak before rising 20% above the trough. 'Confirmed' is the first month in which the 20% decline had occurred, that is, when the bear market became knowable in real time.

4 Methodology

4.1 Markov-switching model of returns

Let $r_t = 100 \cdot \Delta \ln P_t$ be the monthly real return. The main regime model is a two-state Markov-switching autoregression with switching mean and variance (Hamilton, 1989):

$$r_t - \mu_{S_t} = \varphi\big(r_{t-1} - \mu_{S_{t-1}}\big) + \sigma_{S_t}\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \quad S_t \in \{\text{bull}, \text{bear}\}, \tag{1}$$

where S_t follows a first-order Markov chain with transition probabilities $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$. Setting $\varphi = 0$ gives the switching mean-variance model. The bear regime is the one with the lower mean, and the expected duration of regime i is $1/(1 - p_{ii})$. The Hamilton filter delivers the filtered probability

$$Q_t = \Pr(S_t = \text{bear} \mid r_1, ..., r_t), \tag{2}$$

which uses only returns through t and is therefore a legitimate real-time quantity once the parameters are fixed. The smoothed probability $\Pr(S_t = \text{bear} \mid r_1, ..., r_T)$ conditions on the whole sample and is used only for description. An h-step regime forecast is $\Pr(S_{t+h} = \text{bear} \mid r_1, ..., r_t) = [1 - Q_t, Q_t]\boldsymbol{P}^h e_2$, where $\boldsymbol{P}$ is the transition matrix.

4.2 Binary models for dated bear markets

The dated bear indicator y_{t+h} is the forecast target. The binary regression

$$\Pr(y_{t+h} = 1 \mid x_t) = F(\alpha + x_t^\top \beta) \tag{3}$$

uses the logistic (logit) or standard normal (probit) distribution function F. The 6 predictors in x_t are the log dividend yield, the log CAPE ratio with lagged earnings, twelve-month CPI inflation, the twelve-month change in the ten-year Treasury yield, the twelve-month real price change, and twelve-month realized volatility. In-sample estimates use Newey and West (1987) standard errors, since overlapping horizons make the errors serially correlated. Appendix A also reports regressions of the regime probability Q_{t+h} on each predictor, the design of the award version, estimated by OLS and by beta regression (Ferrari and Cribari-Neto, 2004; Smithson and Verkuilen, 2006).

4.3 Real-time design

Forecasts are made at every month-end from January 1960 onward for horizons of 1, 3, 6, 12 months. At each origin t every model is re-estimated on an expanding window using only (i) predictors dated $s \le t$, (ii) vintage bear labels from dating the price history through t, so that a training pair (x_s, y_{s+h}) requires $s + h \le t$, and (iii) for the regime model, parameters re-estimated each January on returns through the origin (64 estimations) and filtered probabilities from those parameters. The training data start in June 1910, when all predictors are first available. I compare 7 forecasts:

- **Unconditional frequency**: the training-sample share of bear months h months ahead.
- **Persistence**: the training-sample frequency of a bear market h months ahead, conditional on the current real-time state (in a confirmed bear market or not).
- **MS-AR(1)**: a logit recalibration of the h-step regime forecast from the filtered probability.
- **Logit: dividend yield**, the single valuation predictor emphasized in the literature.
- **Logit** and **probit** on all 6 predictors.
- **Logit: all predictors + state**, which adds the real-time state to test whether the predictors add to persistence.

Accuracy is measured by the Brier score (Brier, 1950), the log loss (with probabilities truncated to $[0.001, 1 - 0.001]$), the area under the ROC curve (AUC) and calibration curves. Brier skill is the percentage reduction in the Brier score relative to the unconditional frequency. Differences in Brier score are tested with the Diebold and Mariano (1995) statistic with the small-sample correction of Harvey, Leybourne and Newbold (1997) and a Bartlett HAC variance.

4.4 Robustness design

Each robustness experiment changes one design choice and re-runs the affected part of the pipeline. (i) **Regime specification**: two regimes with switching mean and variance with and without the AR(1) term, switching variance only, switching mean only, and three regimes, all on January 1960 to December 2023, plus the main model estimated separately before and after 1990. (ii) **Dating rule**: thresholds of 15% and 25% on real prices and 20% on nominal prices, each followed by a complete real-time forecasting run with re-dated vintage labels and real-time states. (iii) **Estimation window**: rolling 30-year windows instead of expanding ones. (iv) **Evaluation period**: the main forecasts scored on targets before and after 1990 and without the 2007-2009 financial crisis (2007:10 to 2009:06). (v) **Predictors**: 5 additional state variables from the same file, namely the excess CAPE yield, the log payout ratio, twelve-month real earnings growth, the level of the ten-year yield and the drawdown of the real price from its twelve-month high, each as a univariate logit alongside the six main predictors, together with an equal-weight combination of all 11 univariate forecasts (Rapach, Strauss and Zhou, 2010) and a logit on all 11 predictors. (vi) **Sampling uncertainty**: a circular moving-block bootstrap (Künsch, 1989) of target months with 24-month blocks and 2,000 replications, resampling all models jointly, which gives intervals for each Brier skill and for each model's gain over persistence.

5 Results

5.1 Returns have two persistent regimes

Table 3 reports the regime models. In the MS-AR(1) model for January 1960 to December 2023, the bull regime has a mean real return of 0.84% per month with a standard deviation of 2.45%, while the bear regime has a mean of −1.81% and a standard deviation of 5.62%. Both regimes are persistent ($p_{\text{bull,bull}} = 0.944$, $p_{\text{bear,bear}} = 0.808$), implying expected durations of 17.9 months for bull regimes and 5.2 months for bear regimes. The autoregressive term ($\varphi = 0.194$) partly reflects the time averaging of monthly prices and lowers the AIC from 4,026.1 to 3,994.5, so the MS-AR(1) is the main specification. On the longer 1900 to 2023 sample the bull regime is more persistent (expected duration 30.4 months) and the bear regime is more volatile.

Figure 2 contrasts the filtered and smoothed bear-regime probabilities. The smoothed series marks the dated bear markets more cleanly (AUC 0.859 against 0.799 for the contemporaneous dated state), but that advantage comes from conditioning on later returns. The filtered series is noisier, reacting to individual volatile months such as October 1987 and early 2020, and it is the only one a forecaster could have used.

5.2 Inflation, volatility and valuation are associated with bear markets

Table 4 reports full-sample binary models for the dated bear indicator, and Figure 3 plots the logit coefficients. Higher inflation (0.82*** (0.26) at one month) and higher realized volatility (0.51*** (0.18) at six months) raise the probability of a bear market at all horizons. A strong past year lowers the short-horizon probability (−0.78*** (0.25) at one month), which captures persistence: a low twelve-month return usually means a decline is already under way. The dividend yield enters positively (0.97** (0.44) at six months). Because it divides lagged dividends by the current price, it rises when prices fall, so part of this coefficient also reflects an ongoing decline rather than cheapness. The dividend yield and the CAPE ratio are strongly negatively correlated, so their conditional coefficients should not be read separately. The pseudo-R^2 falls from 0.261 at one month to 0.105 at twelve months.

Table 3: Two-regime Markov-switching models of monthly real S&P 500 returns

	MS, 1960-2023	MS-AR(1), 1960-2023	MS, 1900-2023	MS-AR(1), 1900-2023
Mean monthly real return (%)				
Bull regime	0.900^{***}	0.835^{***}	0.773^{***}	0.711^{***}
	(0.131)	(0.144)	(0.097)	(0.117)
Bear regime	-1.887^{***}	-1.812^{***}	-2.221^{***}	-2.231^{***}
	(0.585)	(0.668)	(0.535)	(0.651)
Innovation variance				
Bull regime	6.35	6.01	8.24	7.88
	(0.55)	(0.55)	(0.48)	(0.48)
Bear regime	30.94	31.54	56.48	55.64
	(4.29)	(4.51)	(6.17)	(6.33)
Bull volatility (sd, %)	2.52	2.45	2.87	2.81
Bear volatility (sd, %)	5.56	5.62	7.52	7.46
Autoregression				
AR(1) coefficient		0.194^{***}		0.232^{***}
		(0.035)		(0.026)
Transitions				
P(bull to bull)	0.949	0.944	0.967	0.967
	(0.017)	(0.019)	(0.008)	(0.009)
P(bear to bear)	0.826	0.808	0.860	0.855
	(0.055)	(0.065)	(0.035)	(0.039)
Expected bull duration (months)	19.5	17.9	30.5	30.4
Expected bear duration (months)	5.7	5.2	7.2	6.9
Fit				
Log-likelihood	−2,007.0	−1,990.3	−4,069.3	−4,028.8
AIC	4,026.1	3,994.5	8,150.5	8,071.5
BIC	4,053.9	4,027.0	8,182.4	8,108.7
Observations	768	767	1,486	1,485
Sample	1960:01-2023:12	1960:02-2023:12	1900:03-2023:12	1900:04-2023:12

Notes: Maximum-likelihood estimates; standard errors in parentheses from the numerical Hessian. The bear regime is the regime with the lower mean. Switching mean and variance; the AR(1) coefficient is common to both regimes. Returns are 100 times the log change in the real price index. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided). Stars are shown only for tests against zero that are meaningful (means, AR term).

5.3 In real time, models beat the base rate but not persistence

Table 5 is the central result. At the one-month horizon every model beats the unconditional frequency (Brier 0.198): the multivariate logit reaches 0.157 (skill 20.8%, AUC 0.780) and the probit 0.155. Persistence alone reaches 0.161 (skill 18.9%), and the Diebold-Mariano tests cannot distinguish the multivariate models from persistence ($p = 0.316$ for the logit and 0.256 for the probit). Adding the real-time state to the logit does not help (0.159). The MS-AR(1) forecast ranks months well (AUC 0.702) but is less accurate than persistence (skill 10.9%).

The block bootstrap in Table 6 and Figure 4 puts these differences in perspective. The 90% interval for the one-month skill of persistence is [6.6, 30.7]% and that of the multivariate logit [7.3, 35.4]%; both exclude zero. The logit's gain over persistence, however, ranges from −11.7 to 15.4 percentage points, and the logit beats persistence in a share of 0.61 of the bootstrap draws. With blocks of 12 or 36 months instead of 24 the interval for persistence becomes [7.6, 30.4]% or [7.5, 30.2]%, so the conclusion does not depend on the block length.

5.4 Skill fades with the horizon

At three months the single-variable dividend-yield logit is the most accurate model (Brier 0.175, skill 12.3%), essentially tied with persistence (skill 11.5%, $p = 0.433$), while the multivariate logit falls to 6.2%. At six months only the dividend-yield logit retains a meaningful gain (skill 7.9%, $p =$

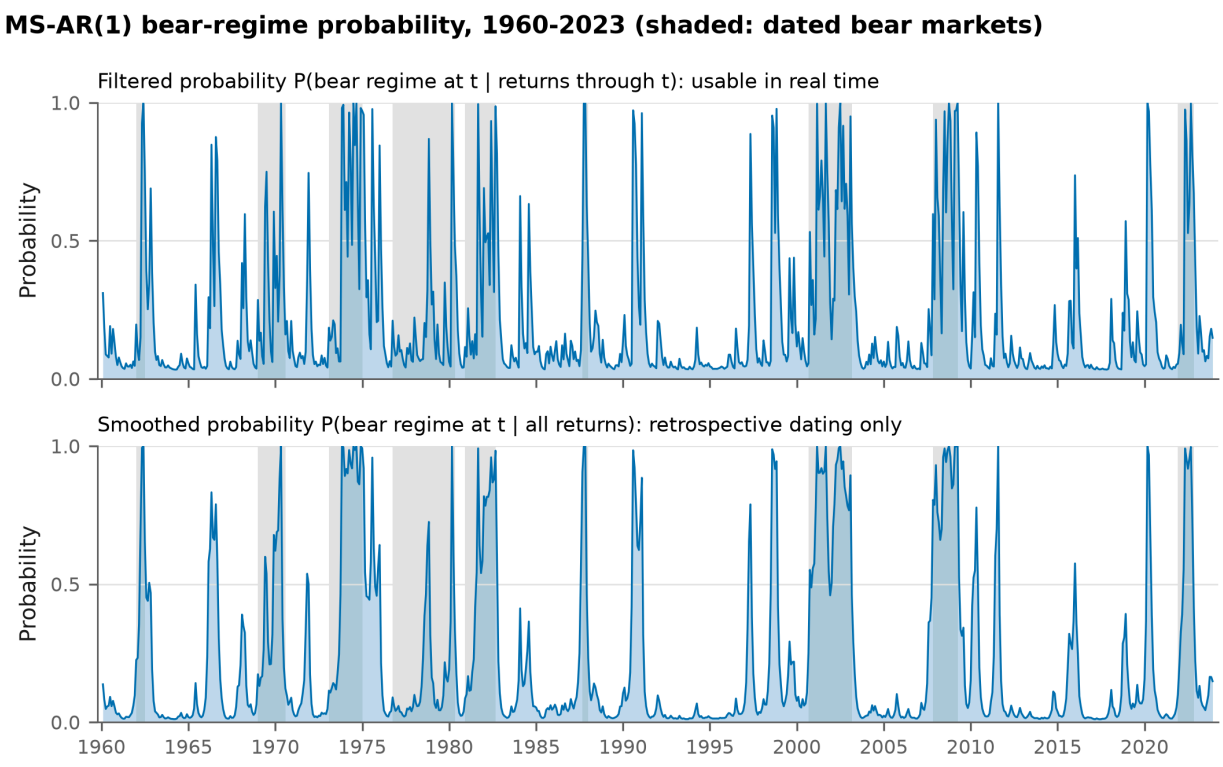


Figure 2: Filtered and smoothed bear-regime probabilities from the MS-AR(1) model, with dated bear markets shaded. Only the filtered probability is used for forecasting.

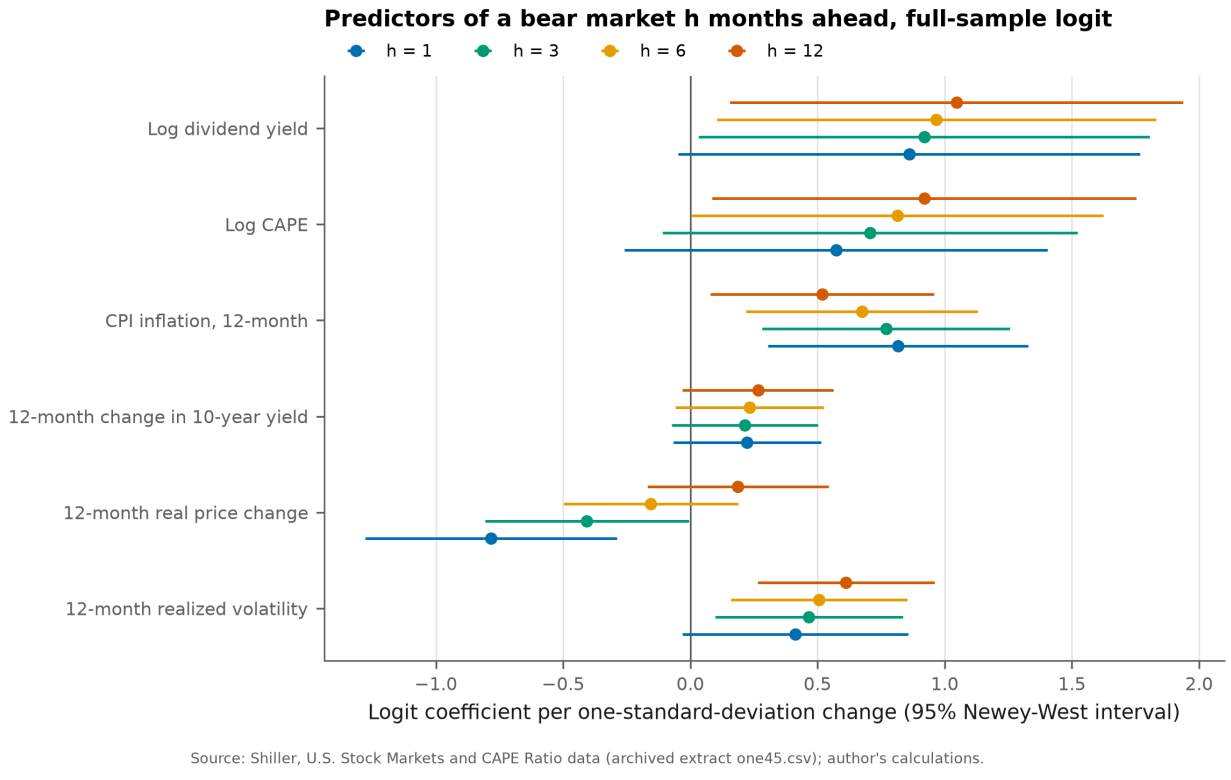
0.051 against the unconditional frequency), and the multivariate models are worse than the base rate (skill −6.9%). At twelve months no model is reliably better than the unconditional frequency: the best skill is 1.9% ($p = 0.353$), and the multivariate logit has an AUC of 0.417, below the 0.5 of an uninformative forecast. Across all 20 comparisons of a model with persistence, none is significant at the 5% level (smallest $p = 0.256$).

Figure 5 shows why. The model probabilities rise sharply once a decline is under way and fall when the market recovers, so at short horizons they track bear markets well, mostly with a lag. In the months before the peaks of 1973, 2000 and 2007 the model probabilities were below the unconditional frequency. The calibration curves in Figure 6 show that the short-horizon forecasts are directionally right but overconfident at the top: months assigned the highest probabilities turned out to be bear months less often than predicted. At twelve months the curves are flat, and the ROC curves in Figure 7 lie close to the diagonal.

Table 4: Full-sample logit and probit models of a dated bear market h months ahead

	Logit				Probit			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
Valuation								
Log dividend yield	0.86*	0.92**	0.97**	1.05**	0.49*	0.53**	0.57**	0.62**
	(0.46)	(0.45)	(0.44)	(0.46)	(0.25)	(0.25)	(0.25)	(0.26)
Log CAPE	0.57	0.71*	0.81**	0.92**	0.32	0.40*	0.48**	0.55**
	(0.42)	(0.42)	(0.41)	(0.43)	(0.23)	(0.24)	(0.24)	(0.25)
Macroeconomic								
CPI inflation, 12-month	0.82***	0.77***	0.67***	0.52**	0.46***	0.43***	0.39***	0.31**
	(0.26)	(0.25)	(0.23)	(0.22)	(0.14)	(0.14)	(0.13)	(0.13)
12-month change in 10-year yield	0.22	0.21	0.23	0.27*	0.14*	0.14	0.14	0.15*
	(0.15)	(0.15)	(0.15)	(0.15)	(0.09)	(0.09)	(0.09)	(0.09)
Market								
12-month real price change	−0.78***	−0.41**	−0.16	0.19	−0.43***	−0.24**	−0.10	0.10
	(0.25)	(0.20)	(0.18)	(0.18)	(0.12)	(0.11)	(0.10)	(0.11)
12-month realized volatility	0.41*	0.47**	0.51***	0.61***	0.25**	0.28***	0.30***	0.37***
	(0.23)	(0.19)	(0.18)	(0.18)	(0.11)	(0.10)	(0.10)	(0.10)
Constant	−0.95***	−0.89***	−0.85***	−0.82***	−0.58***	−0.54***	−0.52***	−0.49***
	(0.18)	(0.17)	(0.18)	(0.19)	(0.10)	(0.10)	(0.10)	(0.11)
Pseudo-R^2 (McFadden)	0.261	0.186	0.137	0.105	0.260	0.186	0.137	0.104
Log-likelihood	−633.3	−697.1	−737.6	−763.0	−634.2	−696.9	−737.4	−763.6
AIC	1,280.7	1,408.3	1,489.1	1,540.1	1,282.4	1,407.9	1,488.8	1,541.2
Observations	1,362	1,360	1,357	1,351	1,362	1,360	1,357	1,351

Notes: Dependent variable: 1 if month t+h lies in a dated bear market. Predictors are measured at t and standardized. Newey-West standard errors in parentheses (9, 9, 9, 12 lags by horizon). Forecast origins 1910:06 to 2023:11 for h = 1. * p < 0.10, ** p < 0.05, *** p < 0.01 (two-sided). In-sample estimates over the whole period, not real-time forecasts.



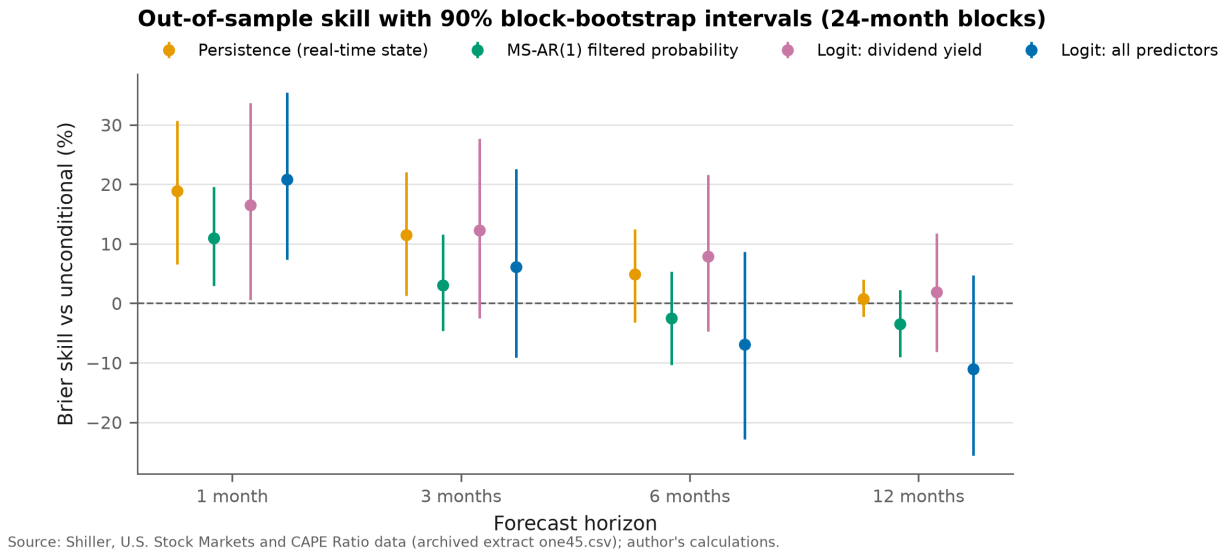
Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 3: Full-sample logit coefficients per one-standard-deviation change in each predictor, with 95% Newey-West confidence intervals, by forecast horizon.

Table 5: Out-of-sample accuracy of real-time bear-market forecasts, targets 1960:02-2023:12

	Brier	Skill (%)	Log loss	AUC	p vs uncond.	p vs persist.
h = 1 (767 forecasts, bear share 22.7%)						
Unconditional frequency	0.198	0.0	0.587	0.563		
Persistence (real-time state)	0.161	18.9	0.502	0.688	<0.001	
MS-AR(1) filtered probability	0.176	10.9	0.537	0.702	<0.001	0.987
Logit: dividend yield	0.165	16.5	0.527	0.627	<0.001	0.680
Logit: all predictors	0.157	20.8	0.488	0.780	<0.001	0.316
Probit: all predictors	0.155	21.5	0.487	0.779	<0.001	0.256
Logit: all predictors + state	0.159	19.9	0.497	0.772	<0.001	0.387
h = 3 (765 forecasts, bear share 22.7%)						
Unconditional frequency	0.199	0.0	0.589	0.547		
Persistence (real-time state)	0.176	11.5	0.537	0.636	<0.001	
MS-AR(1) filtered probability	0.193	3.0	0.575	0.618	0.182	0.994
Logit: dividend yield	0.175	12.3	0.545	0.586	0.001	0.433
Logit: all predictors	0.187	6.2	0.563	0.675	0.134	0.878
Probit: all predictors	0.185	7.0	0.561	0.676	0.099	0.838
Logit: all predictors + state	0.189	5.1	0.573	0.667	0.197	0.933
h = 6 (762 forecasts, bear share 22.8%)						
Unconditional frequency	0.200	0.0	0.592	0.528		
Persistence (real-time state)	0.190	4.9	0.568	0.571	0.094	
MS-AR(1) filtered probability	0.205	−2.5	0.602	0.530	0.737	0.986
Logit: dividend yield	0.185	7.9	0.567	0.541	0.051	0.287
Logit: all predictors	0.214	−6.9	0.623	0.549	0.843	0.981
Probit: all predictors	0.213	−6.3	0.621	0.548	0.826	0.977
Logit: all predictors + state	0.218	−8.6	0.634	0.537	0.879	0.992
h = 12 (756 forecasts, bear share 23.0%)						
Unconditional frequency	0.202	0.0	0.595	0.513		
Persistence (real-time state)	0.201	0.8	0.591	0.513	0.331	
MS-AR(1) filtered probability	0.209	−3.5	0.609	0.463	0.847	0.959
Logit: dividend yield	0.198	1.9	0.594	0.488	0.353	0.405
Logit: all predictors	0.224	−11.1	0.654	0.417	0.911	0.939
Probit: all predictors	0.226	−11.8	0.659	0.411	0.924	0.949
Logit: all predictors + state	0.227	−12.3	0.660	0.407	0.929	0.952

Notes: Expanding-window forecasts made at each month-end origin using only data available then: predictors with publication lags, bear dates obtained from prices through the origin, and Markov-switching parameters re-estimated each January (filtered probabilities only). Lower Brier score and log loss are better; skill is the percentage reduction in Brier score relative to the unconditional frequency. Log loss truncates probabilities to [0.001, 0.999]. The p-values are one-sided Diebold-Mariano tests (Harvey-Leybourne-Newbold correction, Bartlett HAC variance) of equal Brier score against the alternative that the model is more accurate than the baseline.



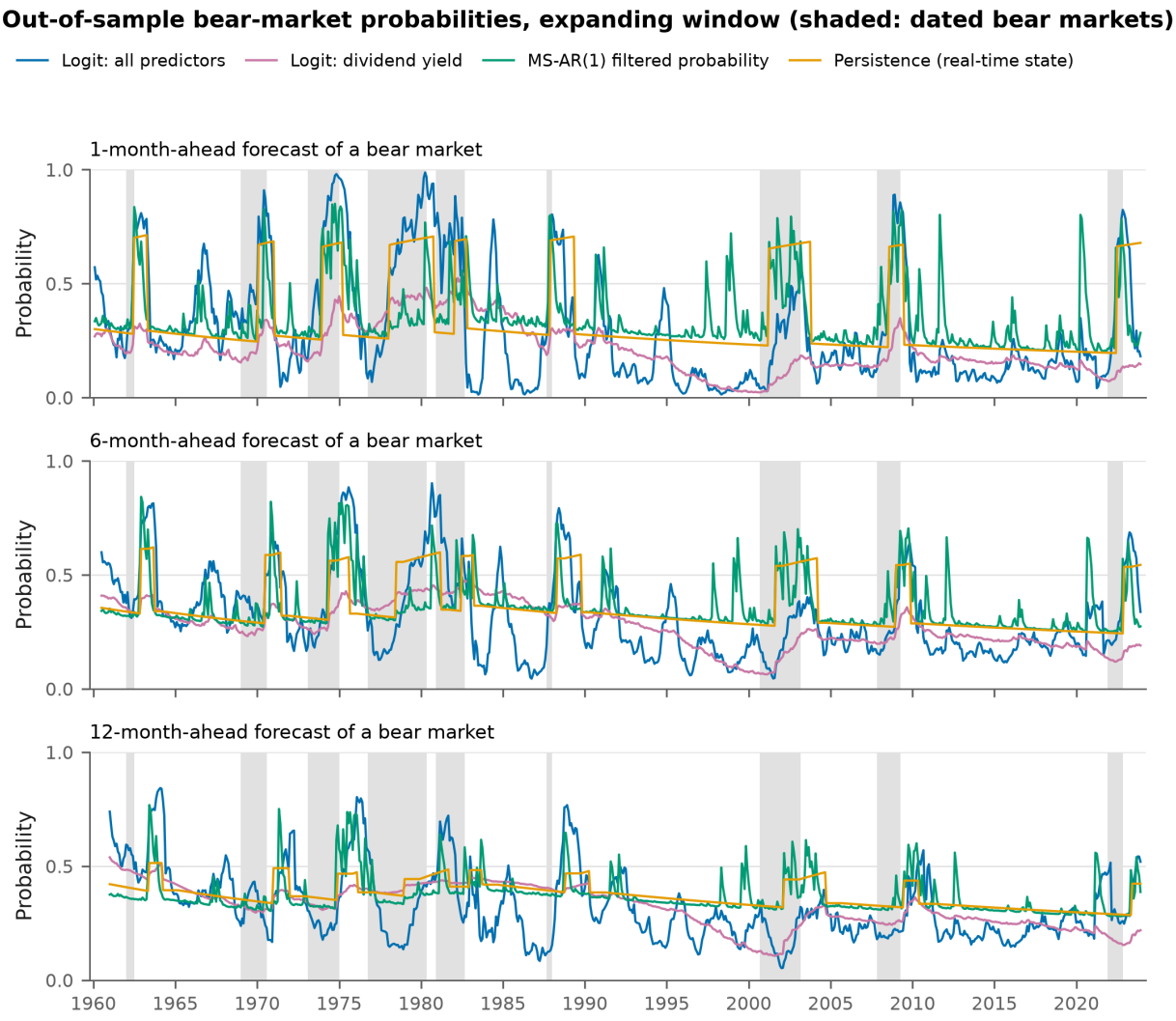
Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 4: Out-of-sample Brier skill relative to the unconditional frequency, with 90% moving-block bootstrap intervals. The dashed line is the unconditional frequency.

Table 6: Block-bootstrap uncertainty of out-of-sample Brier skill

	Skill (%)	90% interval	Gain over persistence (pp), 90% interval	Share of draws beating persistence
h = 1				
Persistence (real-time state)	18.9	[6.6, 30.7]		
MS-AR(1) filtered probability	10.9	[3.0, 19.6]	[−19.6, 3.8]	0.14
Logit: dividend yield	16.5	[0.5, 33.7]	[−20.9, 16.3]	0.45
Logit: all predictors	20.8	[7.3, 35.4]	[−11.7, 15.4]	0.61
Probit: all predictors	21.5	[8.2, 35.8]	[−11.0, 16.1]	0.65
Logit: all predictors + state	19.9	[5.3, 35.2]	[−10.4, 12.6]	0.58
h = 3				
Persistence (real-time state)	11.5	[1.3, 22.1]		
MS-AR(1) filtered probability	3.0	[−4.6, 11.6]	[−17.1, 0.1]	0.05
Logit: dividend yield	12.3	[−2.5, 27.7]	[−16.2, 16.0]	0.55
Logit: all predictors	6.2	[−9.1, 22.6]	[−19.4, 7.8]	0.27
Probit: all predictors	7.0	[−8.0, 23.3]	[−18.6, 8.7]	0.31
Logit: all predictors + state	5.1	[−11.1, 22.1]	[−18.9, 5.8]	0.20
h = 6				
Persistence (real-time state)	4.9	[−3.2, 12.5]		
MS-AR(1) filtered probability	−2.5	[−10.3, 5.3]	[−14.8, −0.5]	0.03
Logit: dividend yield	7.9	[−4.7, 21.6]	[−9.2, 15.3]	0.67
Logit: all predictors	−6.9	[−22.8, 8.7]	[−25.1, 1.5]	0.07
Probit: all predictors	−6.3	[−21.7, 9.2]	[−24.5, 2.0]	0.08
Logit: all predictors + state	−8.6	[−24.8, 7.7]	[−26.3, −0.9]	0.04
h = 12				
Persistence (real-time state)	0.8	[−2.3, 4.0]		
MS-AR(1) filtered probability	−3.5	[−9.0, 2.3]	[−8.2, −0.3]	0.03
Logit: dividend yield	1.9	[−8.1, 11.8]	[−8.0, 9.9]	0.60
Logit: all predictors	−11.1	[−25.6, 4.7]	[−25.2, 2.3]	0.08
Probit: all predictors	−11.8	[−26.6, 4.1]	[−26.1, 1.6]	0.07
Logit: all predictors + state	−12.3	[−27.3, 3.7]	[−26.8, 1.4]	0.07

Notes: Circular moving-block bootstrap of target months with blocks of 24 months and 2,000 replications, resampling all models jointly. Intervals are 5th and 95th percentiles. The gain over persistence is the model's skill minus the skill of persistence in the same draw.



Source: Shiller, U.S. Stock Markets and CAPE Ratio data (archived extract one45.csv); author's calculations.

Figure 5: Real-time forecast probabilities, plotted at the target month, with dated bear markets shaded. Each forecast uses only data available at its origin.

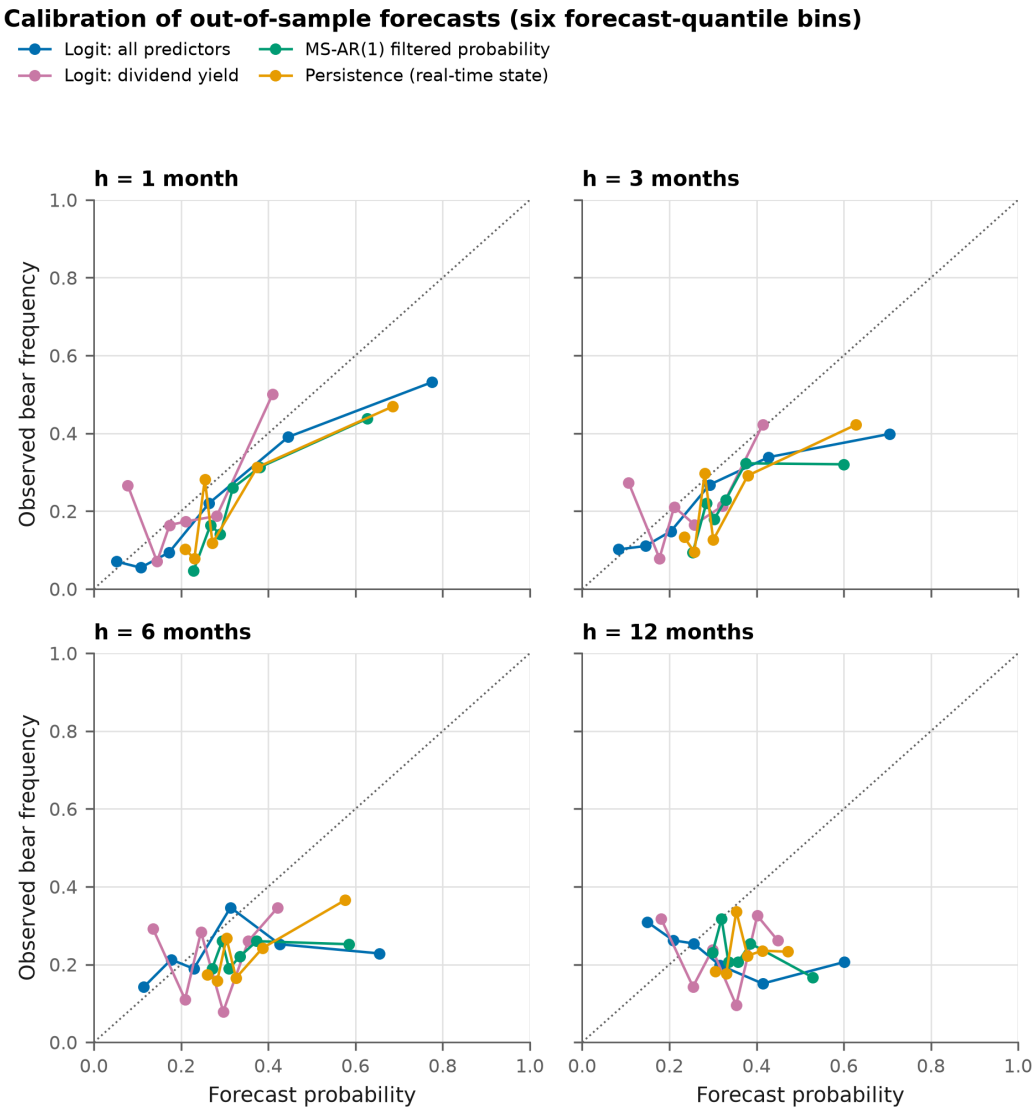


Figure 6: Calibration of out-of-sample forecasts: observed bear frequency against the mean forecast within six forecast-quantile bins. The dotted line is perfect calibration.

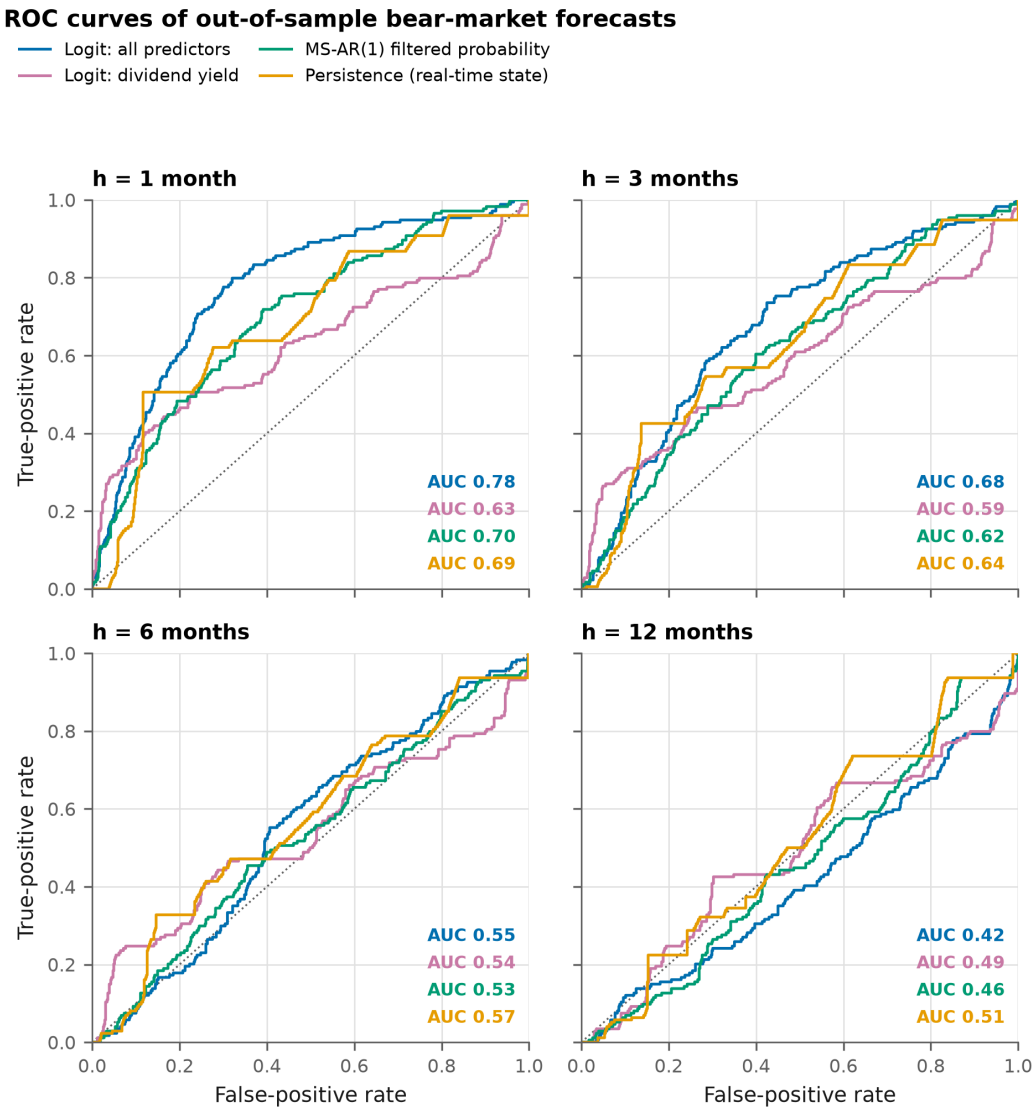


Figure 7: ROC curves of out-of-sample forecasts by horizon, with areas under the curve.

6 Robustness

6.1 Regime specification and sample

Table 7 compares the main model with four alternatives on January 1960 to December 2023 and with the main model estimated on two subsamples. The MS-AR(1) has the lowest AIC (3,994.5) and BIC (4,027.0) in the first group. Letting only the variance switch gives a high-volatility regime (standard deviation 6.15% against 2.56%) that tracks dated bear markets less well (AUC 0.742 against 0.799). Letting only the mean switch picks out short crash months rather than bear markets: the low-mean regime averages −10.05% a month and lasts 1.4 months. A three-regime model splits the bull state into a calm regime (mean 0.66%, duration 18.4 months) and a short rebound regime (mean 3.45%, duration 3.3 months), while its bear regime (mean −2.66%, standard deviation 5.02%) closely resembles the two-regime one and tracks dated bear markets better (AUC

Table 7: Robustness of the regime model: specification and sample

	MS-AR(1)	Mean and variance	Variance only	Mean only	Three regimes	MS-AR(1), 1960-1989	MS-AR(1), 1990-2023
Mean monthly real return (%)							
Bull (highest mean)	0.84	0.90	0.60	0.67	3.45	0.62	1.05
Middle (three regimes)					0.66		
Bear (lowest mean)	−1.81	−1.89	0.60	−10.05	−2.66	−2.45	−1.52
Volatility (sd, %)							
Bull (highest mean)	2.45	2.52	2.56	3.02	2.75	2.57	2.29
Middle (three regimes)					2.37		
Bear (lowest mean)	5.62	5.56	6.15	3.02	5.02	5.63	5.68
Expected duration (months)							
Bull (highest mean)	17.9	19.5	22.7	35.8	3.3	10.8	19.3
Middle (three regimes)					18.4		
Bear (lowest mean)	5.2	5.7	6.4	1.4	6.0	2.4	6.4
Fit							
AR(1) coefficient	0.194					0.242	0.138
Parameters	7	6	5	5	12	7	7
Log-likelihood	−1,990.3	−2,007.0	−2,019.6	−2,027.9	−1,991.6	−935.9	−1,047.3
AIC	3,994.5	4,026.1	4,049.2	4,065.9	4,007.2	1,885.8	2,108.6
BIC	4,027.0	4,053.9	4,072.5	4,089.1	4,062.9	1,912.9	2,136.6
Observations	767	768	768	768	768	359	407
Filtered bear probability against the dated bear state							
AUC	0.799	0.826	0.742	0.769	0.874	0.732	0.899
Correlation	0.42	0.46	0.35	0.30	0.55	0.31	0.58

Notes: Two-regime models unless stated; monthly real returns. The first five columns use 1960-2023; AIC and BIC are comparable only within that group. 'Mean and variance' is the switching mean-variance model without the AR term; 'Variance only' lets only the variance switch (its regimes share one mean and the bear regime is the high-variance one); 'Mean only' lets only the mean switch. In the three-regime model the bear regime is the one with the lowest mean. The last two columns re-estimate the main MS-AR(1) on each subsample. AUC and correlation compare the filtered bear-regime probability with the contemporaneous dated bear indicator (in sample, full-sample parameters).

0.874). Its extra parameters do not pay for themselves by AIC (4,007.2). Across subsamples the bear regime's volatility is stable (5.63% before 1990, 5.68% after), whereas the bull regime became more persistent (expected duration 10.8 months before, 19.3 after) and the filtered probability became a much better real-time marker of bear markets (AUC 0.732 before, 0.899 after).

6.2 Dating rule

Table 8 shows how the set of episodes depends on the rule. A 15% threshold dates 31 bear markets since 1900 and a 25% threshold 17, against 21 for the main rule. Dating nominal prices yields only 7 episodes since 1960, because inflation in the 1970s turned long real declines into flat nominal prices; the bear share falls to 14.1%. Table 9 re-runs the entire real-time exercise under each rule. The pattern is unchanged. Persistence has a one-month skill between 15.8% (nominal) and 24.6% (15%), the dividend-yield logit retains positive skill at one and three months under every rule, and the multivariate logit has negative skill at twelve months under every rule (−9.4%, −9.7% and −15.0%).

Table 8: Bear markets under alternative dating rules

Dating rule	Bear markets, 1900-2023	Since 1960	Bear share since 1960 (%)	Median decline (%)
Real prices, 15%	31	12	24.6	30.0
Real prices, 20% (main)	21	9	22.7	34.8
Real prices, 25%	17	7	24.6	42.4
Nominal prices, 20%	19	7	14.1	32.2

Notes: Symmetric thresholds: a bear market starts when the index falls the stated percentage below its peak and ends when it rises the same percentage above its trough. Nominal dating uses the S&P Composite monthly average price without deflating.

6.3 Estimation window and evaluation period

Rolling 30-year windows reduce the skill of persistence, the regime model and the dividend-yield logit at every horizon (for example 3.7% for the dividend-yield logit at one month, against 16.5%), and the multivariate logit deteriorates sharply at long horizons (−48.3% at twelve months). The longer history in the expanding window is valuable, which is expected for a rare event: thirty years contain only a handful of bear markets.

The evaluation-period rows of Table 9 show that forecastability is concentrated in the later part of the sample. For targets from 1990 onward, persistence reaches a one-month skill of 32.1% and the multivariate logit 33.9%; for earlier targets the corresponding figures are 7.9% and 10.0%. The long real bear markets of 1968-1982 were confirmed slowly, 11 to 16 months after their peaks (Table 2), while the declines of 2000-2002 and 2007-2009 were confirmed within 7 and 9 months. Excluding the 2007-2009 crisis changes little (one-month skill 20.0% for persistence and 22.2% for the logit), so the results are not driven by a single episode.

6.4 Additional predictors and forecast combination

Table 10 evaluates each of the 11 predictors as a univariate logit in the same real-time design. The strongest single predictors at one month are price-based: the drawdown from the twelve-month high (skill 27.7%, AUC 0.844) and the twelve-month real price change (23.7%). The drawdown is the only forecast in this study that beats persistence at the 5% level ($p = 0.008$), and its advantage disappears by three months ($p = 0.158$). It measures how far a decline has already gone before the 20% threshold is crossed, so it recognizes the same ongoing bear markets slightly earlier than the real-time state. The additional valuation and macroeconomic predictors add nothing: the excess CAPE yield, the payout ratio, earnings growth and the level of the ten-year yield all have one-month skills between −4.5% and 0.5%. At six and twelve months the dividend yield remains the best single predictor (7.9% and 1.9%).

The equal-weight combination is stable but unspectacular: its skill is 14.4%, 8.2%, 3.2% and −1.0% at the four horizons: positive up to six months and close to zero at twelve, but below persistence at every horizon. The logit on all 11 predictors overfits as the horizon lengthens (−28.9% at twelve months).

Taken together, the alternative designs and the predictor experiments involve 136 comparisons of a model with persistence, of which 3 are significant at the 5% level, all at the one-month horizon. That is no more than chance alone would produce.

7 Limitations

The sample contains 9 bear markets since 1960, so the out-of-sample evaluation rests on few independent episodes and the overlapping monthly forecasts are strongly dependent. The block bootstrap and the HAC-based tests account for this dependence only approximately, and the wide bootstrap intervals are an honest summary of how little a sample of this size can resolve. Monthly average prices understate intra-month declines: the 2020 decline, for example, does not reach the 20% threshold in monthly averages. Publication lags are imposed conservatively, but the Shiller series are themselves revised and partly interpolated, so this is a quasi-real-time rather than a true vintage exercise. The predictor set is limited to what the archived file contains; credit spreads and the term structure, which Chen (2009) and Chen, Chen and Chou (2017) find useful, are not available in it.

Table 9: Robustness of out-of-sample Brier skill (%) to dating, estimation window and evaluation period

	h = 1	h = 3	h = 6	h = 12
Persistence				
Main (real prices, 20%, expanding)	18.9***	11.5***	4.9*	0.8
Dating threshold 15%	24.6***	13.7***	6.6**	0.1
Dating threshold 25%	17.1***	10.0***	3.6	−1.3
Nominal prices, 20%	15.8***	9.9***	4.8*	0.9
Rolling 30-year window	10.6**	2.9	−3.3	−5.8
Targets 1960-1989	7.9*	2.1	−2.3	−1.8
Targets 1990-2023	32.1***	22.6***	13.5***	3.7*
Excluding 2007-2009 crisis	20.0***	12.6***	6.1*	1.3
MS-AR(1)				
Main (real prices, 20%, expanding)	10.9***	3.0	−2.5	−3.5
Dating threshold 15%	10.1***	2.3	−2.5	−2.6
Dating threshold 25%	11.1***	4.9*	−0.3	−2.6
Nominal prices, 20%	16.9***	8.0**	−0.5	−3.1
Rolling 30-year window	7.6**	−0.7	−4.5	−5.1
Targets 1960-1989	1.9	−6.4	−9.3	−7.7
Targets 1990-2023	21.8***	14.3***	5.6	1.5
Excluding 2007-2009 crisis	10.7***	2.7	−2.2	−3.3
Logit: dividend yield				
Main (real prices, 20%, expanding)	16.5***	12.3***	7.9*	1.9
Dating threshold 15%	12.4***	8.4***	4.6	−2.6
Dating threshold 25%	11.0***	7.8**	3.9	−0.8
Nominal prices, 20%	8.2**	5.2*	1.4	−4.8
Rolling 30-year window	3.7*	0.3	−1.5	−6.4
Targets 1960-1989	11.7***	6.6**	2.4	−2.9
Targets 1990-2023	22.3***	19.1***	14.4*	7.4
Excluding 2007-2009 crisis	19.8***	15.2***	10.4**	3.6
Logit: all predictors				
Main (real prices, 20%, expanding)	20.8***	6.2	−6.9	−11.1
Dating threshold 15%	23.1***	7.3*	−5.4	−9.4
Dating threshold 25%	26.8***	14.8***	2.3	−9.7
Nominal prices, 20%	14.0***	1.0	−11.8	−15.0
Rolling 30-year window	21.0***	−2.1	−24.8	−48.3
Targets 1960-1989	10.0*	−7.0	−21.7	−27.8
Targets 1990-2023	33.9***	21.8***	10.6	8.3
Excluding 2007-2009 crisis	22.2***	7.6*	−4.9	−9.2

Notes: Skill is the percentage reduction in Brier score relative to the unconditional frequency estimated in the same variant. Each dating variant re-dates bear markets (symmetric thresholds) and re-runs the complete real-time exercise, including the vintage labels and the real-time state. The rolling variant estimates every model on the most recent 30 years of training data. The evaluation-period rows score the main forecasts on subsets of target months; the crisis excluded is 2007:10-2009:06. Stars: one-sided Diebold-Mariano test against the unconditional frequency, * p < 0.10, ** p < 0.05, *** p < 0.01.

8 Conclusion

The regime-switching description of the S&P 500 is robust: real returns alternate between a long, calm bull regime and a short, volatile bear regime in every specification and subsample examined here. The claim that market and macroeconomic indicators forecast bear markets is much weaker once forecasts are made in real time. In real time, the information that best predicts a bear market next month is that one has already begun. Models built on inflation, valuation, momentum and volatility do not significantly improve on that persistence benchmark; the price drawdown improves on it only at one month and only by recognizing the decline sooner. Beyond a few months, the dividend yield retains a small and imprecise advantage over the unconditional frequency, and nothing else does. These conclusions survive changes to the dating rule, the regime model, the estimation window, the evaluation period and the predictor set. They are consistent with a market in which downturns are difficult to anticipate from public information, and they suggest that the

Table 10: Out-of-sample Brier skill (%) of individual predictors and combinations

	h = 1	h = 3	h = 6	h = 12	AUC, h = 1
Baseline					
Persistence	18.9***	11.5***	4.9*	0.8	0.688
Univariate logits: main predictors					
Log dividend yield	16.5***	12.3***	7.9*	1.9	0.627
Log CAPE	14.4***	9.0***	3.3	−5.1	0.653
CPI inflation, 12-month	10.0***	7.8***	5.0***	1.3*	0.769
12-month change in 10-year yield	2.0	0.6	−0.2	0.2	0.655
12-month real price change	23.7***	12.6***	4.5*	−0.5	0.821
12-month realized volatility	2.5	−0.1	−2.4	−2.3	0.593
Univariate logits: additional predictors					
Excess CAPE yield	0.5	−3.1	−6.6	−12.9	0.543
Log payout ratio	−0.5	−1.5	−2.3	−3.0	0.415
12-month real earnings growth	0.0	−0.9	−1.3	−1.2	0.617
10-year Treasury yield	−4.5	−6.0	−7.6	−8.1	0.548
Drawdown from 12-month high	27.7***	14.9***	4.3	−1.1	0.844
Multi-predictor forecasts					
Equal-weight combination (11 univariate logits)	14.4***	8.2***	3.2**	−1.0	0.767
Logit: all 11 predictors	24.7***	5.7	−14.5	−28.9	0.779

Notes: Real-time expanding-window forecasts with the main dating rule; every model is re-estimated at each origin on data available then. The combination averages the probability forecasts of the univariate logits with equal weights. Skill is relative to the unconditional frequency. Stars: one-sided Diebold-Mariano test against the unconditional frequency, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

useful contribution of these models is faster recognition of a bear market, not advance warning of one.

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A Regressions of the regime probability and the award version

The award version of this thesis forecast the filtered regime probability Q_t rather than a dated event, with univariate OLS and beta regressions of Q_{t+h} on each of 7 monthly-change predictors,

$$Q_{t+h} = \alpha + \beta x_t + e_{t+h}, \tag{4}$$

and an out-of-sample comparison with an intercept-only model. Table 11 re-estimates this design with predictors observable at t, growth rates in place of index levels, and Newey-West standard errors. The number of predictors significant at the 5% level is 4 of 7 at one month, 3 at three, 6 at six and 2 at twelve. The largest R^2 (0.121) belongs to the real total return at one month (coefficient -9.20^{***} (1.12) percentage points), which is close to mechanical because the filter itself updates on recent returns. Dividend growth (-3.80^{***} (1.33) at one month) and inflation have the expected signs; total-return-scaled earnings show no relationship ($R^2 = 0.001$). Because Q_t uses full-sample parameters, these are in-sample associations rather than forecasts.

Table 13 reports the out-of-sample comparison. Panel C uses expanding-window refits from April 2011: 16 of 28 OLS ratios of mean squared prediction errors fall below one and 13 are significant at the 10% level by the Clark and West (2007) test, with a best ratio of 0.865. The regime probability's own current value achieves 0.476 at one month and no gain at six months (1.032).

The award version reached stronger conclusions for three reasons that a replication of its code path confirms (Table 12): its regime model was estimated with the following month's price as a regressor, its "k months ahead" targets were shifted backward rather than forward in time, and its loss ratio compared squared mean errors rather than mean squared errors. The replication reproduces the reported transition probabilities (0.950 and 0.833 against 0.945 and 0.833) and the

Table 11: Corrected archived specification: regime probability h months ahead on each predictor

	OLS (percentage points)				Beta regression (logit link)			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
CPI inflation, monthly	3.643**	4.065*	4.512**	4.267***	0.155**	0.200**	0.180**	0.171***
	(1.824)	(2.282)	(1.760)	(1.530)	(0.076)	(0.095)	(0.071)	(0.061)
R^2 / pseudo-R^2	0.019	0.024	0.029	0.026	0.020	0.027	0.029	0.030
Change in 10-year yield	−0.066	−1.349	3.575***	0.944	0.017	−0.052	0.131***	0.036
	(1.329)	(1.100)	(1.026)	(0.815)	(0.060)	(0.040)	(0.043)	(0.028)
R^2 / pseudo-R^2	0.000	0.003	0.018	0.001	0.000	0.003	0.017	0.001
Real bond return	−0.104	1.060	−3.911***	−1.386	−0.018	0.039	−0.145***	−0.052*
	(1.412)	(1.303)	(1.212)	(0.909)	(0.060)	(0.048)	(0.049)	(0.031)
R^2 / pseudo-R^2	0.000	0.002	0.022	0.003	0.000	0.001	0.021	0.003
Real dividend growth	−3.805***	−3.337**	−3.395**	−2.808	−0.122***	−0.112**	−0.115**	−0.111
	(1.329)	(1.344)	(1.388)	(2.034)	(0.047)	(0.044)	(0.047)	(0.073)
R^2 / pseudo-R^2	0.021	0.016	0.017	0.011	0.017	0.014	0.015	0.015
Real total return	−9.195***	−6.111***	−5.763***	−1.306	−0.411***	−0.241***	−0.218***	−0.039
	(1.118)	(1.099)	(1.295)	(0.981)	(0.045)	(0.047)	(0.052)	(0.032)
R^2 / pseudo-R^2	0.121	0.053	0.048	0.002	0.107	0.049	0.043	0.002
TR-scaled earnings growth	−0.859	−1.595	−2.083	−2.630**	−0.082	−0.065	−0.066	−0.085*
	(2.498)	(1.782)	(1.660)	(1.309)	(0.123)	(0.073)	(0.062)	(0.049)
R^2 / pseudo-R^2	0.001	0.004	0.006	0.010	0.003	0.003	0.005	0.007
Change in log TR-CAPE	−9.128***	−6.109***	−5.849***	−1.507	−0.407***	−0.242***	−0.222***	−0.046
	(1.115)	(1.116)	(1.289)	(1.011)	(0.044)	(0.047)	(0.052)	(0.033)
R^2 / pseudo-R^2	0.120	0.053	0.049	0.003	0.106	0.049	0.045	0.002
Observations	766	764	761	755	766	764	761	755

Notes: Each cell is a separate univariate regression of the filtered bear-regime probability Q(t+h) from the MS-AR(1) model on the predictor at t, which is standardized (coefficients are per one standard deviation). OLS coefficients are in percentage points of probability. Newey-West standard errors in parentheses (7, 12 lags by horizon). Forecast origins 1960:02 to 2023:11 for h = 1; the sample shortens by h months. The beta-regression pseudo-R^2 is the squared correlation of the linear predictor with logit(Q). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided). Q is computed with full-sample parameters, so these are in-sample associations, not forecasts.

Table 12: Archived results, their replication, and the corrected estimates

	Archived (reported)	Replicated	Corrected
P(bull to bull)	0.945	0.950	0.944
P(bear to bear)	0.833	0.833	0.808
Dividend, k = 1: coefficient	−0.195	−0.198	−3.80*** (1.33) pp
Dividend, k = 6: coefficient	−0.325	−0.320	−3.40** (1.39) pp
Dividend, k = 6: R^2	0.075	0.074	0.017
Total return, k = 1: R^2	0.108	0.107	0.121
TR-scaled earnings, k = 1: R^2	0.085	0.085	0.001
Dividend, k = 1: OLS forecast ratio	0.004	0.029	0.953
Total return, k = 1: OLS forecast ratio	0.014	0.006	0.868
TR-scaled earnings, k = 1: OLS forecast ratio	0.390	0.363	0.999

Notes: Archived values are transcribed from the award PDF (v2/data/archived_reported.csv). 'Replicated' re-runs the archived code path in Python, including the future-price regressor, the backward target Q(t-k) and the archived MSPE formula. 'Corrected' uses the MS-AR(1) model on returns, past-only predictors, the forward target Q(t+k) and expanding-window forecasts. Coefficients in the corrected column are per one-standard-deviation change of a different (growth-rate) transformation and are therefore comparable in sign and significance only.

reported coefficients (correlation 1.000). Panels A and B of Table 13 show the archived design with its own ratio and with the correct one.

Table 13: Forecasting the regime probability: unrestricted versus intercept-only models

	OLS				Beta regression			
	h = 1	h = 3	h = 6	h = 12	h = 1	h = 3	h = 6	h = 12
A. Archived design and archived formula (squared mean error ratio)								
CPI	0.923	0.735	0.868	0.925	0.947	0.904	0.961	0.993
Long-term rate	0.981	0.975	1.008	0.997	0.992	0.981	1.002	0.996
Real bond	0.818	0.862	1.077	0.948	0.960	0.949	1.014	0.959
Real dividend	0.029	0.189	0.404	0.179	0.370	0.275	0.210	0.276
Real total return	0.006	0.921	0.942	0.982	0.564	1.019	0.956	1.021
TR-scaled earnings	0.363	0.474	0.759	1.365	0.819	0.837	0.894	1.222
TR-CAPE	0.822	1.016	1.018	1.027	0.706	1.010	1.004	1.013
B. Archived design, correct mean squared prediction error ratio								
CPI	1.002	1.021	1.005	1.006	0.996	0.996	0.997	1.000
Long-term rate	0.989	1.009	1.002	1.001	0.991	1.002	1.001	1.002
Real bond	0.997	1.041	1.019	1.011	0.970	1.027	1.006	1.015
Real dividend	1.154	1.184	1.148	0.963	0.951	0.924	0.883	0.806
Real total return	1.539	1.010	1.003	1.002	0.770	1.001	1.000	0.999
TR-scaled earnings	0.862	0.879	0.946	0.965	0.799	0.861	0.945	1.003
TR-CAPE	0.720	0.994	0.998	0.990	0.664	0.996	1.000	0.994
C. Corrected design: past-only predictors, true h-month lead, expanding window								
CPI inflation, monthly	1.003	0.951^{**}	0.918^{**}	0.939^{***}	0.976^{*}	0.930^{***}	0.920^{***}	0.935^{***}
Change in 10-year yield	1.002	1.000	0.985	1.005	1.004	0.998	0.990	1.003
Real bond return	1.007	1.003	0.987	1.012	1.008	0.997	0.999	1.010
Real dividend growth	0.953^{***}	0.965^{**}	0.935^{***}	0.941^{**}	0.934^{***}	0.943^{***}	0.930^{***}	0.935^{***}
Real total return	0.868^{***}	0.931^{***}	0.985^{*}	1.005	0.844^{***}	0.936^{***}	0.974^{**}	1.000
TR-scaled earnings growth	0.999	1.007	1.029	1.056	0.997	1.004	1.013	1.025
Change in log TR-CAPE	0.865^{**}	0.928^{***}	0.983^{**}	1.009	0.851^{***}	0.939^{***}	0.978^{**}	1.003
Benchmark: regime probability at t	0.476^{***}	0.872^{***}	1.032	1.025	0.483^{***}	0.853^{***}	0.985	1.004

Notes: Entries are ratios of the unrestricted model's loss to the intercept-only model's loss; values below one favor the predictor. Panels A and B re-run the archived code path (target Q(t-k), leaked regime model, single 80/20 split with the test period starting in 2011). Panel A uses the archived formula (mean error)² / (mean error)², which is not a mean squared error. Panel C forecasts Q(t+h) from predictors observable at t with expanding-window refits, origins 2011:04 to 2023:11 for h = 1. The benchmark row uses Q(t) itself as the predictor. Stars in Panel C mark Clark-West (2007) tests of equal accuracy in nested models where the unrestricted model is better: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.